Both workflows on one card

Build, check, hand on. Sixteen steps, fifteen pieces of work: schematic 8 and layout 1 are one crossing.

S Schematic: what connects to what

1 Create the project

DONE WHEN

The folder holds .kicad_pro, .kicad_sch and .kicad_pcb, and the schematic is open.

L1.1

2 Place symbols

DONE WHEN

Every part, and every power connection, has a symbol on the sheet.

L1.2

3 Annotate

DONE WHEN

No part still shows a question mark: no R? or U? left.

L1.3

4 Set component values

DONE WHEN

No part that needs a value still reads the bare R or C it started with.

L1.4

5 Wire and label

DONE WHEN

Every pin's unconnected-end square has gone, and the signals you care about carry labels.

L1.5

6 Assign footprints

DONE WHEN

Every symbol on the sheet has a footprint.

L1.6

7 CHECK Run the ERC

DONE WHEN

Zero violations, or every one read and judged harmless. Fix and re-run until it is.

L1.7

8 Hand off to the board

DONE WHEN

Save, then Update PCB from Schematic (F8). Every part and connection has landed on the board.

L1.8

Update PCB from Schematic (F8): the one-way crossing

Changing what connects? Back to the schematic, save, cross again. Never on the board.

1 Import from the schematic

DONE WHEN

The same command as schematic 8. Every part is on the canvas and the ratsnest is showing.

L1.8

2 Draw the board outline

DONE WHEN

One closed outline on Edge.Cuts surrounds the area where the parts will sit.

L1.10

3 Place components

DONE WHEN

The ratsnest is as short and calm as you can make it, and no courtyards overlap.

L1.11

4 Route the tracks

DONE WHEN

Every connection you route by hand is copper. Only the ground lines wait for the fill.

L1.12

5 Add copper fills

DONE WHEN

The zone is drawn on GND and filled. Press B to refill after any change.

L1.13

6 CHECK Run the DRC

DONE WHEN

Zero violations, or every one read and judged harmless. Fix and re-run until it is.

L1.14

7 Add text and silkscreen

DONE WHEN

The board carries its name and version. Run the DRC once more.

L1.15

8 Generate the Gerbers

DONE WHEN

3D view checked. One Gerber per layer you need, plus the drill file.

L1.16

L Layout: where everything goes

THE ORDER HOLDS

Footprints and a clean ERC before you cross
The copper fill before the DRC
A clean DRC before the Gerbers
Change what connects in the schematic, then cross again

THE ORDER BENDS

Annotate and set values in either order
Move the outline when the parts need room
Silkscreen can come early: re-run the DRC after

Lost? Find the first step whose "done when" is not true yet: that is where you are. Each tag names the Module 1 lecture that shows the step.